

Shared evidence | Bt crop trial board

Genetic engineering: change, test, decide

Name _____ Date _____

This is a paper evaluation. Crop-trial numbers are constructed sample data, not observations from a named commercial product.

Mechanism
Some Bt crops contain added genetic information originating from *Bacillus thuringiensis*. The crop can produce a protein that harms particular insect pests. It does not become a bacterium and is not protected from every pest.

Evidence limits
The trial covers one site and one season. It reports damage and yield, not food safety or effects on every non-target species. Neither crop received target-insect spray. All applications need evidence relevant to the crop, trait and setting.

Possible resistance risk
Pests vary. Inherited variants that survive exposure may reproduce, so resistance can become more common over generations. Monitoring and resistance-management strategies aim to delay this.

Crop	Target-pest damaged plants /100	Mean grain yield (t/ha)
Comparison non-GM	30	6.9
Bt GM	8	7.9

Constructed sample data: means from three matched plot pairs. Water, fertiliser and planting density were matched. One site; one season.

W1-W2. What does the trial show, and which claim goes beyond it?

W3. Choose two next checks: repeat sites/seasons; monitor resistance; measure crop colour; count likes online. Justify them.

Supported route | Build your recommendation

Use the shared evidence. You may speak, annotate or dictate. Words: DNA • protein • target pest • evidence • resistance • test.

S1. Finish: genetic engineering directly alters A Bt gene can help the crop make

S2. Choose one trial comparison. Include both numbers and what they measure.

S3. Explain one benefit for this crop in this trial. Do not say it proves all GM organisms are the same.

S4. Finish: resistance may become more common because surviving pests

S5. Recommend a next step. Name one question and a test that could answer it.

Standard route | Evidence-based trial recommendation

Write for a grower who needs a clear reasoned next step, not a slogan.

M1. Explain how genetic engineering differs from selective breeding. Include the direct DNA change.

M2. Explain how an added Bt gene can change crop performance. Connect DNA, protein and a target-pest effect.

M3. Compare damage and yield using the constructed trial values. Identify one benefit that is supported here.

M4. Explain a plausible risk using a biological mechanism. Identify one question the table does not answer.

M5. Give your provisional recommendation and a specific next test. Explain what result might change your view.

Stretch route | Evaluate and extend

Choose T1-T3 for deeper evaluation. T4 is an optional Higher-tier mechanism extension; it is not required for the Foundation core.

Constructed sample data: original trial mean yield non-GM6.9 versus Bt7.9 t/ha; target-pest damage30 versus8 plants per100. A second season at another site gives non-GM7.1 versus Bt7.0 t/ha and damage5 versus4 plants per100. Different pest pressure and growing conditions may matter.

T1. Compare both yield differences and damage differences. What does the second setting change about your conclusion?

T2. Explain why neither trial alone establishes food safety or all environmental effects. Propose one focused additional investigation.

T3. State a provisional recommendation that considers a benefit, a risk, a missing piece of evidence and who might be affected.

Optional Higher-tier mechanism box: in one classical method, a restriction enzyme cuts DNA at particular sequences. Compatible short single-stranded ends (“sticky ends”) can pair. Ligase joins the DNA backbone. A vector, such as a plasmid, carries the DNA into a suitable cell. This is one conceptual method, not all genetic engineering.

T4. Explain the different jobs of the restriction enzyme, complementary sticky ends, ligase and vector. Why does inserting DNA not guarantee a useful product?

Exit and reflection | A precise judgement

Answer independently. A short oral or annotated answer is valid.

E1. Define genetic engineering in one sentence.

E2. Give one measured benefit from the constructed trial: non-GM damage 30/100 and yield 6.9 t/ha; Bt damage 8/100 and yield 7.9 t/ha.

E3. Give one plausible risk and a useful next check.

L1. What audience question sharpened your recommendation?

L2. Write one question to take into the Week 4 GM research and digital slideshow.

My next useful step is...

- A. Explain the DNA-to-protein link.
- B. Read a comparison with its units.
- C. Explain resistance across generations.
- D. Match an uncertainty to a test.